

Andres Perez

+1 (562) 333-5244 | aperez26@nd.edu | [linkedin.com/in/andres-perez0](https://www.linkedin.com/in/andres-perez0) | github.com/andres-perez0

EDUCATION

University of Notre Dame

Bachelor of Science in Computer Engineering | Minor: Engineering Corporate Practice

Activities: Irish Sat (CubeSat) Team, Career Assistant Intern

Engineering Study Abroad: London, England

Graduation Date: May 2028

GPA: 3.713/4.0

May. 2025 – Jun. 2025

RESEARCH EXPERIENCE

University of Notre Dame

Department of Electrical Engineering – Student Researcher

Notre Dame, IN

Aug. 2025 – Present

- Developing a real-time Python software interface for tracking and visualizing GPS-enabled test resources, like drones and software-defined radios, that parses NMEA strings to extract GPS data and displays it on a live map using the folium library, supporting various research experiments.

Department of Computer Science and Engineering - Student Researcher

Aug. 2024 – Present

- Researched under Prof. Chaoli Wang by data labeling image data, downloading GSV data of households, and surveying heat dissipation in household.

ENGINEERING PROJECTS

Hardware-Accelerated Human Activity Classifier

Long Beach, CA

- Built a wireless data acquisition system using two Arduino Uno R3s, an MPU-9250 IMU sensor, and nRF24L01 RF transceivers.
- Developed a C++ sketch to sample raw accelerometer and gyroscope data, format it, and transmit it wirelessly to a receiving station. Along with a Python script to receive, parse, and log the real-time sensor data, generating a custom labeled dataset for human activities like walking and sitting.
- Trained a 1D Convolutional Neural Network in PyTorch to classify human activities (walking, sitting), achieving a peak validation accuracy of 96.7% on the test set.
- Visualized training and testing accuracy against learning rate, window size, and optimizer to demonstrate model stability and performance trends, documenting the findings.

IrishSat's Pi Hat PCB

Notre Dame, IN

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino, magnetometers, and H-bridge circuits, reducing assembly complexity.
- Amended the generated Bill of Materials and Component Placement List files export to JLCPCB for manufacturing.

Air Quality Sensor Housing | Engineering Design

Notre Dame, IN

- Collaborated with a team of four engineering students to develop and iterate on sensor housing with a rack-and-pinion opening mechanism actuated by a servo motor and ESP32-based microcontroller.
- Designed and prototyped with SolidWorks in two design phases, optimizing for form, fit, function, and durability through iterative testing and feedback.

Portfolio Website | andres-perez.github.io

Long Beach, CA

- Developed an student and engineering project portfolio website with Astro and Tailwind CSS frameworks as the front end.

LEADERSHIP AND ACTIVITIES

IrishSat's Gravitational Orbital Attitude Thermal Lab

Notre Dame, IN

Electronics Lead

Aug. 2024 – Present

- Appointed as incoming Electronics Lead to oversee embedded systems development and PCB design initiatives for the Helmholtz cage project, coordinating hardware-software integration efforts across the cubesat projects.
- Engineered a Python-based data acquisition pipeline for magnetometer readings from Arduino, converting raw serial outputs into MATLAB-compatible formats for post-processing and modeling
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.

TECHNICAL AND LANGUAGE SKILLS

Programming: C, C++, Python, Git, MATLAB, Astro

Tools: Microsoft Office Suite, KiCad, Vim, SolidWorks, Fusion 360

Skills: Technical Writing, Electronics Soldering

Language: English (Native), Spanish (Bilingual)

Hobbies: Japanese(N4-N5), Chess (1000+ Rapid)